JULY 15 - 19, 2024

RISE
RESEARCH & INNOVATION FOR THE SERENGETI ECOSYSTEM

GRUMETI FUND

# ADVANCED GIS TRAINING

FOR EARLY TO MID-CAREER CONSERVATION PRACTITIONERS

**HOSTED BY**
**RISE AT GRUMETI FUND**

CREDIT: DANIEL SAMWEL

# A WORD FROM THE ORGANISERS

Grumeti fund through the Research and Innovation for the Serengeti Ecosystem (RISE) welcomes you to this training on advanced GIS. RISE seeks to bring together the brightest minds from Tanzania and globally, from graduate students to early career scientists, to work collaboratively on research projects that are fundamentally linked to key questions in conservation science and to produce solutions that are simultaneously innovative and practical.

RISE is committed to building in-country capacity and investing in passionate, talented people who are dedicated to the conservation and management of Tanzania's world-renowned protected areas and natural resources.

This year, RISE has launched a new training program in collaboration with experts from Penn State University, the University of Connecticut, the University of Dar es salaam and the Montana States University Billings. Advanced GIS Training (AGT-2024) will be our first program to include international participants.

We are excited to introduce our talented participants from more than 10 countries. We hope this training will help many people advance their careers and add more geospatial skills in their institutions and organizations. For those who got an opportunity to attend this training in-person, we hope you will enjoy your stay and experienc e the wonders of Serengeti.

**DR GEORGE LOHAY**

HEAD SCIENTIST
RISE-GRUMETI FUND

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# MEET THE TEAM

## PARTNERS BEHIND THE ADVANCED GIS TRAINING

**DR. DONALD AKANGA**

**VICTORIA MKESSA**

**DR. OLIPA SIMON**

**DR. GEORGE LOHAY**

**DR. DAN WANYAMA**

**DR. SUSAN KOTIKOT**

**DR. GRACE MALLEY**

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Credit: Daniel Samwel

# ABOUT THE TRAINING PROGRAMME

The Advanced GIS Training is organized by collaborators from RISE, Pennsylvania State University, Montana State University Billings, the University of Connecticut, and the University of Dar es Salaam. 10 candidates have been selected for in-person participation at the RISE facility at Grumeti, Tanzania, and 10 for online participation.

The training will delve into advanced topics in GIS, mainly covering applications of important geospatial methods in the analysis of climate and environmental data. This training is designed for professionals, researchers, and enthusiasts who have some basic background in GIS and would like to deepen their knowledge of geospatial data analysis and applications in analyzing environmental issues.

The training will center on the use of advanced geospatial analysis methods to analyze freely available climate and environmental data. Modules to be covered include GIS fundamentals for ecologists, landscape pattern analysis in R, spatial analysis of ecological data in R, spatial data sources and access, landscape analysis in R and Google Earth Engine (GEE), and habitat connectivity.

A free ArcGIS Pro license valid for one year will be provided to each participant who does not have access to the software. However, participants should bring their own laptops during the training.

# OUR TRAINING OBJECTIVES

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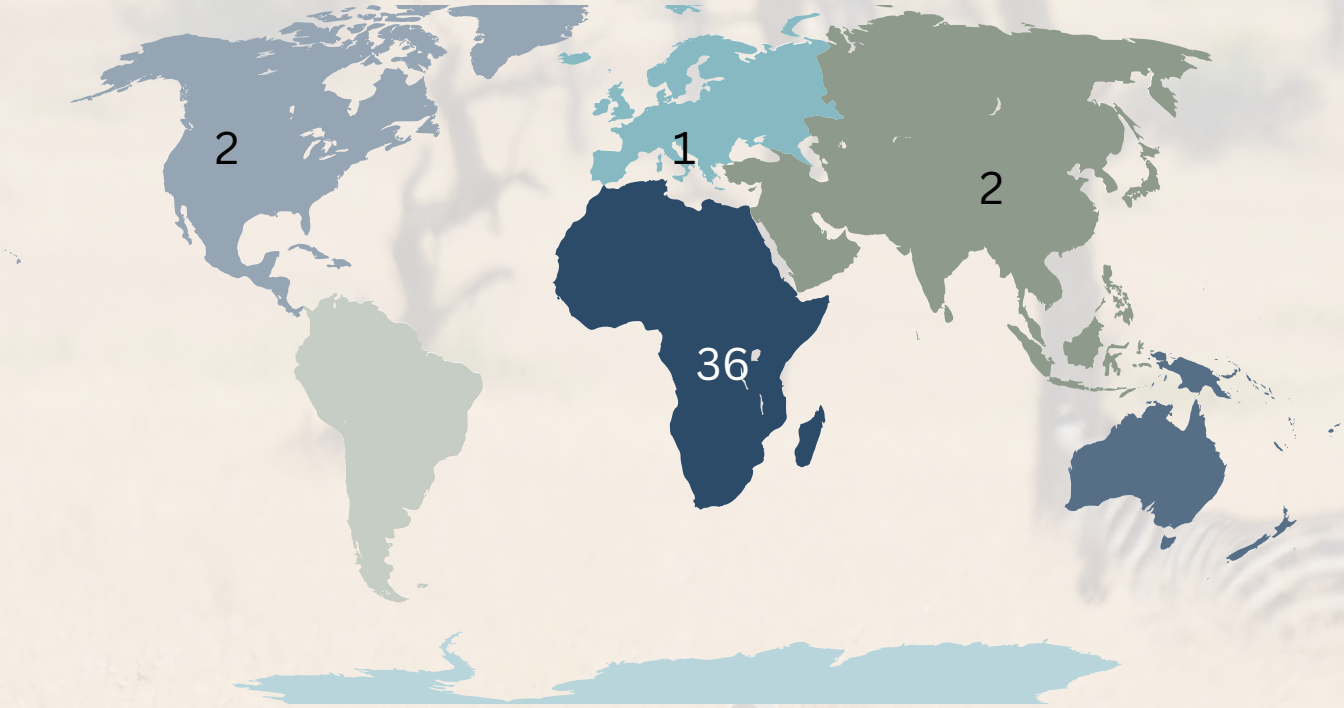
- Identify, download, and process open-source climate and environmental data.
- Perform analysis of climate, environmental, and ecological data in commercial GIS software (ArcGIS Pro) and open-source software.
- Apply GIS and spatial statistics to analyze landscape structure and pattern.
- Apply GIS and multi-criteria analysis techniques to map habitat connectivity for selected species.
- Apply time series analysis to assess patterns and changes in important landscape characteristics and functions.
- Present, in the form of maps and figures, results from their geospatial analysis.

# DELIVERY FORMAT

A typical training day will consist of three sections: self-learning, group work, and live sessions. There will be two instructors on-site and three will deliver their materials remotely. Participants are expected to take charge of their own learning experience.

Credit: Daniel Samwel




# A GLOBAL VIEW OF OUR COHORT

Our cohort is quite diverse. We have representatives from Tanzania (20), Kenya (8), Rwanda (2), Zambia (1), Zimbabwe (1), Botswana (1), Nigeria (1), India (2), Democratic Republic of Congo (1), South Africa (1), England (1) and the United States of America (2).

# INTRODUCING OUR INSTRUCTORS

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## Dr. Dan Wanyama

Assistant Professor, University of Connecticut

Dr. Wanyama earned his PhD in Geography and Geospatial Science from Michigan State University in 2021, an MS in Geospatial Science from University of North Alabama in 2017, and a BS in Environmental Studies and Community Development from Kenyatta University in 2013. His research focuses on the application of geospatial technologies and data to study the shifting human-environment interactions associated with climate change, especially in agricultural lands and forests both in Eastern Africa and the United States. Before joining UConn, Dr. Wanyama was a Post-doctoral Research Associate at the University of Oklahoma, mainly working to integrate data from multiple satellites to study human-associated and nature-driven disturbance and recovery regimes in tropical forests of Africa.

## Dr. Olipa Simon

Leading Scientist, University of Dar es salaam

Dr. Olipa Simon is a leading scientist in Geographic Information Systems (GIS) and Remote Sensing with a distinguished career at the University of Dar es Salaam (UDSM), Tanzania. Her PhD research on the impact of urbanization on land surface temperature exemplifies her focus on using these technologies for environmental problem-solving. Dr. Olipa's academic achievements, including a Master's in Computer Science and a First Class Honors BSc in Land Surveying (Geomatics), are matched by her extensive experience. With over 20 years at UDSM, she now heads the GIS Laboratory, trains students, consults on projects, and actively contributes to national initiatives like Tanzania's Status of Environmental report. Her expertise is further recognized by awards and involvement in high-profile projects, solidifying her position as a leader in advancing GIS and Remote Sensing for Tanzania's natural resource management

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## Dr. Susan Kotikot

Post-doctoral Fellow, Penn State University

Dr. Susan Kotikot is a Postdoctoral Research Associate in the Earth and Environmental systems institute at the Pennsylvania State University, mostly working on land use modeling for dryland agropastoral landscapes, and partly on dynamics of mangrove ecosystems. Dr. Kotikot earned her PhD in Geography and Climate Science from the Pennsylvania State University in 2023, an MS in Earth System science from University of Alabama Huntsville in 2017, and a BS in Environmental Planning and Management from Kenyatta University in 2013. She is a transdisciplinary researcher with broad interests in landscape ecology, climate change, remote sensing, and GIS. Her research mostly focuses on understanding landscape change, and the collective role of climate and human activities with the goal of promoting social-ecological resilience.

## Dr. Don Akanga

Assistant Professor, Montana State University Billings

Dr. Akanga holds a Ph.D. in Geography and a Graduate Certificate in Spatial Ecology from Michigan State University, an MS in Geospatial Science from University of North Alabama, and BS in Environmental Studies and Community Development from Kenyatta University. Some of the GIS-focused courses that Dr. Akanga teaches include Mapping Techniques, Principles of GIS, and Applied GIS/Spatial Analysis. Dr. Akanga mainly employs GIS, remote sensing, and quantitative techniques to study (1) land change science: evaluating patterns, processes, and associated consequences; (2) climate change and socio-ecological sustainability; and (3) livelihoods adaptation and resilience in the context of climate change in East Africa and the United States. He is also a member of the Water Faculty in Montana, USA and has research interests in water resources and sustainable development.

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## Dr. Grace Malley

Lecturer, University of Dar es salaam, Tanzania

Dr. Grace Malley is a lecturer at the Institute of Resource Assessment, University of Dar es Salaam. She holds a Ph.D. in Geography and Environmental Studies from Pennsylvania State University (U.S). Dr Malley holds a Master's degree in Natural Resource Assessment and a BA in Geography and Environmental studies from the University of Dar es salaam. She is an expert in geospatial analysis, landscape ecology, conservation, human-wildlife interactions and social-ecological systems.

Her research involves a comprehensive assessment of persistent changes in land uses and land covers (LULC), and an analysis of spatial-temporal patterns of climatic change in the area – using earth observation and model data. By identifying possible factors driving LULC change and analyzing its past trend, she can model a possible future trajectory to determine where the changes may occur in the future.

## RISE SUPPORT TEAM

From Left to right, Victoria Mkessa (RISE program Coordinator) George Lohay (Head Scientist), Teddy John (Chef) and Emmy John (facility maintenance)

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## CHRISTINA MAPUNDA

Undergraduate Student, CAWM, Tanzania

Christina is an undergraduate student at the College of African Wildlife Management, Mweka. Her interest in wildlife began as president of Tembo Club in school, advocating against poaching by educating students on the ecological and touristic importance of elephants. She specializes in GIS and remote sensing. An aspiring GIS expert, she is dedicated to community service, having volunteered for a year at a charity school. Christina enjoys yoga, jogging, hiking, nature walks and meeting new people.

*"This training will significantly advance my skills in data analysis, visualization, and management of large datasets, particularly in the context of wildlife conservation, while providing hands-on experience with advanced GIS tools like R, Google Earth Engine, and ArcGIS Pro."*

## Athuman Mathayo

MSc. Student, Ecology, UDSM, Tanzania

Athuman holding a BSc in Wildlife Management from the College of African Wildlife Management, Mweka and currently doing a MSc. in Wildlife Ecology at the University of Dar es Salaam, studying elephant demography around Mkomazi National Park. He a committed conservationist and wildlife researcher, with strong experience in community conservation projects, research and GIS mapping.

*"The knowledge and skills that I will be equipped with from this training, will lead me towards best ecosystems conservation practices with the aid of technology."*

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## Serah Muiruri

### Student, Geomatics, Kenya

Serah is currently pursuing a degree in Geomatics Engineering and Geospatial Information Systems. She is deeply passionate about application of Geospatial and Remote Sensing technologies in earth observation, and consequently, in attaining sustainable development goals. I look forward to learning GIS based solutions and engaging with like-minded professionals to gain more knowledge and helping my community in the long run.

*"This advanced training will help me learn how to manipulate remote sensing and spatial data in climate change mitigation and biodiversity in our environment."*

## Kaiza Kaganzi

### Research Assistant, WCPEP, Tanzania

Kaiza has expertise in conservation field and research study design stems from his collaborative endeavors working with both governmental and nongovernmental entities .

*"This training will help me improve my GIS and Remote sensing Skills including but not limited to temporal and spatial data acquisition, temporal and spatial data analysis skills, NDVI data analysis, Familiarize with the ArcGIS pro software since I have only used QGIS which a free software"*

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## Emily Jefferson

### Ph.D. Candidate, Stony Brook, USA

Emily is a current PhD student in the Interdepartmental Doctoral Program in Anthropological Sciences (IDPAS) at Stony Brook University. She is a behavioral ecologist interested in studying how ecological factors shape the experiences of individuals and how those individual experiences in turn influence group-level properties. Specifically, she aims to better understand the evolutionary origins and advantages of living in groups with complex social structure of wild savannah baboons.

*"The goals of my dissertation are to better understand how changes in the ecological environment influence the costs and benefits of group living and this training will help me to understand spatial analyses used for quantifying that habitat change."*

## Chanda Mulenga

### MSc. Student, Geo-information Science, Zambia

Chanda Mulenga holds a BSc. Natural Resources Management and currently pursuing a MSc. Geo-information science and earth observation. He works as an Impact Monitoring officer for REDD+ project in Zambia, utilizing GIS technology. He aspires to become a carbon account.

*"I aspire to become a carbon account, and skills from this program will help me acquire relevant competencies to apply in conservation efforts"*

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## Ofuya Ebenezer Eyituoyo

### Ph.D. Candidate, Akure, Nigeria

Ofuya holds a B. Agric. Tech. in Ecotourism and Wildlife Management and a M. Agric. Tech degree in Biodiversity Management from the Federal University of Technology Akure and now a Ph.D. student at the same University. Ebenezer's Ph.D. research is on "Population Status, Distribution and Threats to the Drill Monkey (Mandrillus leucophaeus) in Cross River National Park, Nigeria".

"*I strongly believe this GIS training will provide me with powerful set of tools and techniques for advance spatial analysis, data management, and visualization,*

## Akshita Dinesh Rabdiya

### Ph.D. Student - Dar es Salaam, Tanzania

Akshita is an ornithologist immersed in the captivating world of birds. Currently pursuing my PhD on the Lesser Flamingo at Lake Natron, she delve into the intricate realms of behavior ecology and migratory patterns. Fascinated by avian life, she spent years observing and documenting their habits and movements. From the fluttering wings of sunbirds to the majestic flights of eagles, every aspect of avifauna enchants me. With a heart full of wonder and a mind eager for discovery, she is dedicated to unraveling the mysteries of our feathered friends and advocating for their conservation.

*"This training will train me for landscape ecology using GIS which further help in analysing my PhD data "*

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## Vallary Awuor Onyando

### Geospatial Eng, University of Nairobi, Kenya

Vallary  holds a Bachelor degree in Geospatial Engineering from the University of Nairobi. She is passionate and keen on applying her skills in achieving the United Nations' Sustainable Development Goals. She has applied her GIS skills in various ways; from mapping striga weed prevalence to generating soil erosion hotspot maps. She is open to exploring the power of geospatial intelligence in solving today's challenges.

*"I would like to explore the use of open source platforms such as R and Google Earth Engine in spatially analysing ecological data."*

## Lloyd Mutungwazi

### Research Technician, Zimbabwe

Lloyd is passionate about conservation technology. He is eager to see how these emerging technologies, can be used to safeguard our fauna and flora. In 2022, he joined African Parks in Matusadona, a non-profit dedicated to biodiversity conservation. As a Research Technician, he participated in fieldworks and gained hands-on experience in wildlife monitoring using some of the tools.

*"The training program will enhance my skills and facilitate a deeper understanding of advanced spatial data analysis"*

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## Zacharia Ambrose Wambura

### Protected Areas and EIA coordinator, Tanzania

Zacharia has worked in natural resources conservation for approximately 20 years, with notable achievements including the successful establishment of several Game Reserves such as Piti, Uwanda, Wami Mbiki, Kilombero, and Luganzo Ntongwe Game Reserve. As a member of the Technical Working Group for the East African Community, he participated in assessing the state of EAC Protected Areas. Currently, he is involved in upgrading protected areas from lower to higher conservation statuses.

*"The training will add more knowledge and skills on mapping protected areas in Tanzania"*

## Noris Mushi

### GIS Analyst, Jane Goodall Inst., Tanzania

Noris holds a degree in Geoinformatics and has dedicated the past decade in advancing her skills geospatial analysis. She has played a pivotal role in integrating geospatial technologies in different sectors including utilities, conservation, and statistics, driving significant improvements in these fields. She is passionate about conservation and aspires to be among leading women in the field, using her expertise to enhance climate change resilience in communities. Her work is marked by a commitment to sustainability and innovation, aiming to make a lasting positive impact on both the environment and society.

*"This training will improve my skillset and provide in-depth knowledge and practical experience in using geospatial tools and techniques for effective and timely decision making."*

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## Ardgard Essau Mwamgeni

MSc. Student, University of Dar es Salaam

Ardgard completed his undergraduate degree in Wildlife science and conservation at the University of Dar es Salaam. He spent more than two years studying and collecting data on the amphibians of the Eastern Arc Mountains particularly, the Uluguru and Mkingu Nature forest reserves. He is currently studying MSc—wildlife ecology at the same University where he will be working on the ecology and conservation of two critically endangered frog species from Tanzania.

*"The advanced training will improve my skills, particularly in understanding GIS in R, the most popular programming language."*

## Jean Jules Cyubahiro

Research Assistant, Museum of Environment, Rwanda

Jean received his bachelor's degree in science with honors in biology from the University of Rwanda in 2022, with the option to study botany and conservation. He completed short online courses on ecosystem restoration offered by UNDP and the training on biodiversity data management offered by the Center of Excellence on Biodiversity and Natural Resource Management.

*"I'll explore GIS functionalities: data analysis methods and spatial principles."*

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## Sila Mbise

Ecologist, TANAPA, Tanzania

My work involves studying and preserving the park's unique ecosystem, focusing on wildlife conservation and ecological research. With a passion for protecting natural habitats, I contribute to various conservation projects in collaboration with Jane Goodall Institute, monitoring endangered species populations in the park and promoting sustainable practices. My experience and commitment to ecological integrity have made a significant impact on the preservation efforts in Gombe National Park, ensuring the protection of its diverse flora and fauna for future generations.

## Finias Fidelis Mwesige

Assistant Lecturer, MJNUAT, Tanzania

Mr. Finias Mwesige, is a graduate of BSc in Agronomy from Sokoine University of Agriculture (SUA) in 2017 and MSc in Natural Resource Assessment and Management (NARAM) from University of Dar es Salaam (UDSM). Mr. Finias is currently serving an academic role in the Department of Agricultural Science at Mwalimu Julius K. Nyerere University of Agriculture and Technology (MJNUAT), also pursuing MSc in Soil Science and Land Management at Sokoine University of Agriculture (SUA) in Tanzania.

“This training will provide me with advancement in valuable skills, especially multi-criteria analysis skills that will help me generate, using GIS, information that can be used as decision-support tools in the fields of agriculture, natural resource conservation, and other related fields.”

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## Caton Schutte

### MSc Candidate, South Africa

An enthusiastic early career biologist with a deep love of nature, stemming from childhood visits to Kruger National Park. Value time spent in the field collecting information that can ground truth analysis. Committed to lifelong learning, and using GIS and R skills to develop data-driven solutions that protect wildlife and foster sustainable coexistence between humans and nature. When not focused on work, playing games, hiking, and spending time with my chocolate labrador, Mylo bring me great joy.

*"Focusing on applying spatial statistics to analyse landscape structure, patterns, and habitat connectivity will improve my skills in assessing wildlife habitat usage and identifying current and potential human-wildlife conflict hotspots."*

## Nyasatu Mkaka

### Intern, Grumeti Fund, Tanzania

Nyasatu Mkaka holds a diploma in International Wildlife Conservation Practice University of Oxford with honors-Robertson WildCRU Award for academic excellence. With three years of conservation work and volunteering. She is currently an intern at Grumeti Fund-RISE, skilled in AI, focusing on camera trap image identification, classifying, and researching large carnivore habitat use, with a particular emphasis on human-wildlife conflict eager to learn GIS for distribution modeling.

*"Participating in GIS advanced training will refine my skills in utilizing geospatial techniques for species modeling and distribution analysis, thus enabling me to contribute more effectively to conservation efforts"*

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## Sarah Maagi

### Intern, Grumeti Fund, Tanzania

Sarah Maagi is an early career Wildlife conservationist with an interest of monitoring wildlife species and their habitats using GIS. She graduated from University of Dar es Salaam with Bachelor's of Science in Wildlife Science and Conservation. Currently, She is an intern with the Department of Research and Monitoring at Grumeti Fund.

*"I hope to gain skills on using ArcGIS software in doing spatial analyses."*

## Njue David Murimi

### Project Officer, POWERCOM, Kenya

David holds a MSc on Environmental Science-(Agro-forestry and Rural Development and a bachelor's degree on Environmental Studies (Community Development). His passion is on agricultural research, natural resources management and Geo-sciences. In the last five years, David has supported farmers improve on their productivity in the changing climatic conditions using modern farming technologies. My research work has yielded publication on peer-reviewed journals that promote suitable farming technologies with a potential to improve yields and conserve natural resources.

"*The training will be an eye-opener on the limitless opportunities that exist in Geosciences, the field I would like to pursue in near future*"

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## David Ruto Ndiwa

### Wildlife & Enterprise Management expert, Kenya

David holds a Bachelor's degree in Wildlife and Enterprise Management, which has provided him with a strong foundation in wildlife management, conservation, and enterprise. With over a year of experience as a manager in the hospitality and hotel industry, as well as four years of academic and practical expertise in enterprise management and wildlife conservation, He has honed his skills and developed a diverse set of abilities. Additionally, Mr. Ndiwa has technical skills in graphic design, data analysis and visualization, digital marketing, and sales.

*"Acquiring skills in using ArcGIS Pro, R, and Google Earth Engine for climate, environmental, and ecological data analysis will broaden my technical expertise and increase my versatility as a conservation practitioner."*

## Sharifu Juma

### Intern, the Maasai Steppe Carnivore Conservation Trust, Tanzania

Sharifu holds a degree in Wildlife Management and is currently an intern at the Maasai Steppe Carnivore Conservation Trust. He plays a crucial role in monitoring large carnivores in Tarangire National Park, leveraging his academic background and practical experience to contribute to wildlife conservation. Sharifu's dedication to preserving predator species helps maintain the park's ecological balance, showcasing his commitment to the field of wildlife management and conservation in one of Tanzania's most significant national parks.

*"The training will equip me with necessary skills for effectively managing and interpreting environmental data."*

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## Meline Klopfenstein

### MSc. Global Biodiversity, Sussex

I have a Msc in Global Biodiversity Conservation from the University of Sussex. I am currently training to become a field guide in Southern Africa to enhance my practical knowledge of the field. I am also preparing for a PhD looking at the elephant cognition which will include monitoring aspects of their ranging behaviour and environment, with the aim of improving conservation management of elephant populations. I have also worked with different organisations to raise awareness on environmental matters.

*"This training will enable me to enhance my GIS knowledge and be better prepared for my future PhD. "*

## Humphrey Haule

### GIS Analyst, TAWA & FZS, Tanzania

Humphrey holds a diploma in wildlife Management from the African College of Wildlife Management-Mweka. He has over 10 years of work experience with the Tanzania Wildlife Authority (TAWA). Mr. Haule is currently working in Selous Game Reserve with the Frankfurt Zoological Society. He helps in making maps and GIS related work.

*"This training at Grumeti Fund will equip me with the skills and expertise necessary to achieve my goals"*

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## Marcel Ngabikwiye

### Research Assistant, Diane Fossey, Rwanda

Marcel is a field biologist in biodiversity research and conservation in Rwanda. Since 2021, He has been a research assistant at the Dian Fossey Gorilla Fund, focusing on the social behaviors of endangered golden monkeys in Volcanoes National Park, a species he studied during his undergraduate research at the University of Rwanda in biology department. He also teaches and supervises undergraduate students during their field courses. I published my BSc project in African Primates Journal and contributed to other scientific publications.

"*This training will boost my understanding to analyze geospatial data using different techniques acquired through learning advanced GIS, Google earth engine and R software and their application in environmental and ecological context.*"

## Lewis Kuria

Student, JKUAT, Kenya

Lewis P. Kuria is a Geospatial Information Scientist currently pursuing his Bachelor's degree at JKUAT. He possesses a strong foundation in geospatial data analysis, remote sensing, and programming, gained through his coursework and various trainings. He is proficient in various GIS software like ArcGIS, QGIS, and ERDAS, and coding languages like R and Python. He is a self-motivated individual with excellent communication and teamwork skills, actively seeking opportunities to leverage his expertise in a professional setting.

*"This training will equip me with advanced skills and knowledge to tackle complex geospatial challenges by further developing my expertise in spatial data analysis and remote sensing, to solve climatic and environmental problems."*

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## Gwamaka Jwan

### MSc. Student, CAWM. Tanzania

Gwamaka is a graduate student with a passion for wildlife conservation, pursuing an MSc. in African Wildlife Ecology and Conservation through a collaboration between Manchester Metropolitan University and the CAWM- Mweka. Recognizing the increasing role of technology in conservation, he has developed a keen interest in GIS and remote sensing, aiming to become a specialist in this field. Driven by a desire to make a real impact, he aspires to harness the power of advanced science for the future of wildlife conservation.

*"The training will equip me to analyze and utilize spatial data for more effective wildlife conservation strategies."*

## Margaret Poulos

### Ph.D. Student, Stanford, USA

Maggie is a PhD student in the Emmett Interdisciplinary Program in Environment and Resource at Stanford University in the United States. Lying at the intersection of marine socioecology and adaptive ocean governance, Maggie is drawn to the puzzle of how to capture an intangible value or worldview for marine protected area management. She is interested in how biocultural approaches are mechanisms for the inclusion of localized social and cultural values in marine planning. Through a co-production of knowledge framework and participatory field research tools, Maggie aims to co-create applied research that makes marine policy a more diverse and equitable space for local and Indigenous communities.

*"This GIS training will give me the next step of skills to feel confident in analyzing coastal and ocean environments for policy and management purposes."*

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## Linda Mugure Kimotho

### GIS Specialist, Kenya

Linda is a GIS specialist with 9 years' experience in Wildlife Conservation. Born and raised with the wonderful view of the undulating Aberdare ranges she gained a deep connection with nature. She has an insatiable passion to understand and protect our intricate yet valuable ecosystems through use of Geospatial technology. She hopes to use her skills to contribute towards development of solutions to current biodiversity loss challenges and influence positive action for the environment.

*"I will be able to use GIS technology to understand and analyze interaction between species and our dynamic landscapes."*

## Bakari Mtili

### PhD. Candidate, Botswana

Bakari Mtili is a dedicated conservationist and PhD candidate at the Okavango Research Institute, Botswana, specializing in Rangeland Ecology and Connectivity. He holds a MSc. in Conservation Management from the University of Glasgow and a BSc. in Wildlife Management from the College of African Wildlife Management, Mweka. His research focuses on utilizing GIS and spatial analysis to map habitat connectivity in the rangelands of Zimbabwe and Botswana. Additionally, Bakari works as an Assistant Lecturer at the College of African Wildlife Management, Mweka, where he shares his expertise with the next generation of conservationists.

*"I am aiming to learn more on different ways of creating resistance layers for connectivity mapping."*

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## Lollian Kosyando

### MSc. Student, University of Glasgow

Lollian is an early career conservationist with interest in movement ecology and animal behavior. She is a graduate student from the University of Glasgow, Conservation Management of African Ecosystems Program. Her passion lies within learning animal behavior particularly elephants at the moment and the intricate dynamics of human wildlife interactions. She is driven by a commitment to protect biodiversity and aim to contribute to the development and implementation of effective conservation strategies that promote harmonious coexistence between people and wildlife.

*"I aim to enhance and refine my skills in analyzing animal movement data and map animal habitat suitability using advanced GIS techniques."*

## Lynn Fanikiso

### GIS Data Technician, WildCRU, Botswana

Lynn Fanikiso is a GIS Data technician under the Trans-Kalahari Predator Programme which is a Botswana extension of the Wildlife Conservation Research Unit (WildCRU). She specializes in satellite, earth observation, remote sensing and GIS. Her academic foundation is in Earth and Environmental Sciences and has completed her masters in Geomatics through the University of Cape Town. Lynn is particularly interested in the interplay of conservation technology and community science. This includes being an advocate for open access data and platforms.

*" ..I believe this training can better my analyst skills towards landscape and climate indicator modelling more so its automation through R and python..."*

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## Juma Minya

Ecologist, TANAPA, Tanzania

Juma Minya has a broad interest in conservation, population biology and movement ecology. He graduated from Sokoine University of Agriculture (TZ) with a BSc. in Wildlife Management, and later employed by Tanzania National Parks (TANAPA) as a park ecologist before completing his MSc. Conservation Management of African Ecosystems at the University of Glasgow (UK). He was a research fellow working on elephant demography study in Grumeti - Ikorongo Game Reserve.

*"..as an ecologist, this advanced GIS training is perfect fit for me and will provide an incredible opportunity to broaden my understanding and improve the techniques to address conservation challenges."*

## Isabel Julia Rowles

Biomonitoring Manager, Limpopo, South Africa

I have a BSc (Hons) Biology from the University of Leeds with a Year in Industry. Since graduating in 2016, I've worked in various private, government and community-owned game reserves across South Africa, including 3 years on an agricultural farm, developing a grazing management strategy and mapping the farm fence lines, and implementing a drone spraying program for alien plant clearing. I enjoy a mixture of fieldwork and office-based work. My major interests are in vegetation assessments (particularly grasses), mapping with QGIS, data management, and data analysis. I am known for attention to detail, and maintaining clean, accurate datasets.

*".. his course will help me develop my skillset, and especially once I obtain my drone pilot license, so this course would compliment both my existing skillset and interests, as well as my professional development plans."*

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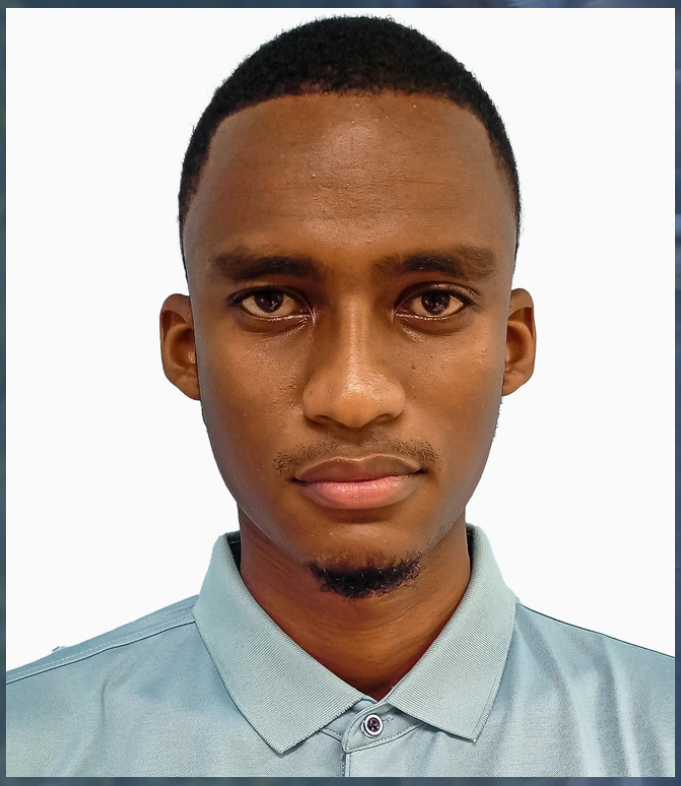
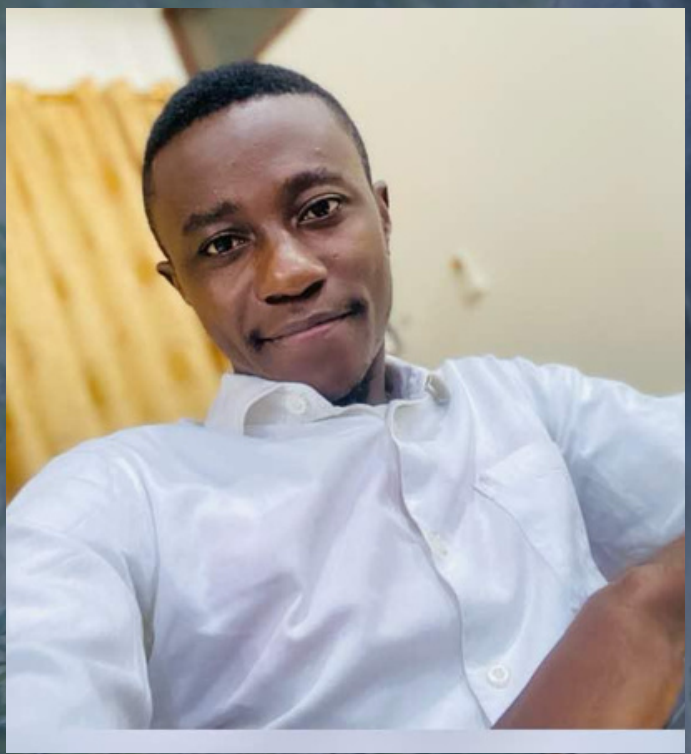


## Suleyman Majaliwa Kyonda

### MSc. Student, Climate change, Zanzibar

Suleyman Kyonda is-an early carrier specialising in big earth data, geospatial analysis, natural resources management and climate change. Holding a M.Sc. in Climate Change and Natural Resources Management and a B.A. in Geography and Environmental Studies from the State University of Zanzibar, Suleyman has extensive experience in GIS, remote sensing, and community mapping. His research includes geospatial data collection, LULC mapping, and climate change adaptation strategies. He has worked as a Climate Change Lead Officer, research assistant, and intern in various capacities, contributing to significant projects and conferences. Suleyman's technical skills include ArcGIS, QGIS, and SPSS.

"*This training will deepen my expertise in advanced GIS techniques, particularly in analysing climate and ecological data, thereby enhancing my ability to conduct comprehensive geospatial analyses for environmental conservation.*"



## Mukaku Gloire

### PhD. Candidate, DRC

Mukaku Gloire is a PhD student at Lubumbashi University. He is also teaching practical aspects of GIS at Kindu University. Mukaku holds a master's degree in integrated planning and management of Forests and Tropical Territories from the Regional postgraduate school and a diploma in Agricultural engineering at the University of Kindu. Currently, he needs to acquire sufficient skills in climate and environmental data acquisition for his doctoral thesis.

*"These skills will not only benefit my own academic journey but will also have a positive impact on my community. As a mentor to hundreds of students at my university, I will be able to impart this knowledge and contribute to their employability in the job market."*

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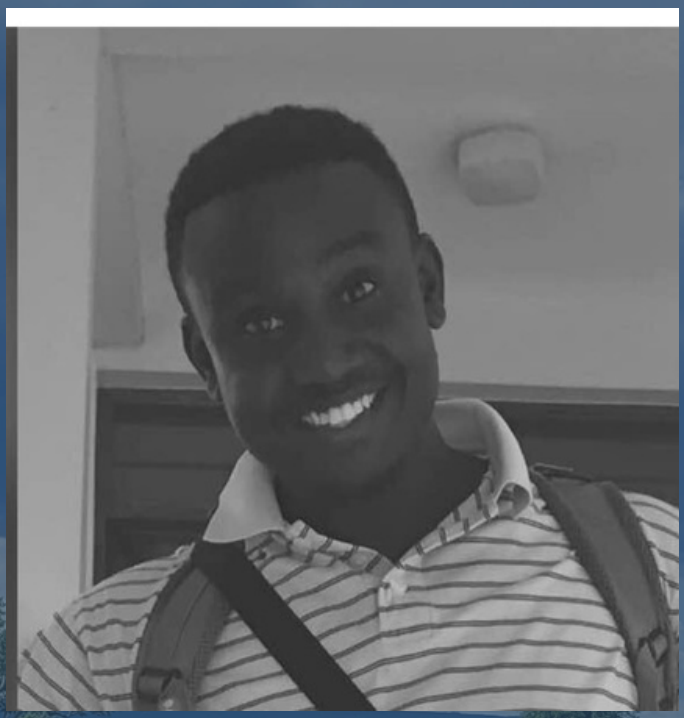

## Gerald Mafuru

### Senior GIS Officer, TANAPA, Tanzania

Gerald Mafuru is a senior GIS officer with TANAPA. He is currently a master's student of Geographical information Science at the University of Dar es salaam. He also holds a B.A in Geography and Environmental Studies from the University of Dar es salaam. Gerald's work has been instrumental within TANAPA as he has helped improve data collection through GIS platforms.

*"I believe this training presents an excellent opportunity for me to deepen my knowledge and skills in advanced geospatial analysis methods". Gerald is looking for Ph.D. scholarship opportunities"*

## Elizabeth Wanderi

### Fisheries Officer & Ph.D. Student, Eldoret, Kenya

Elizabeth is an enthusiastic fisheries officer working for Kenya Fisheries Service, stationed at the largest permanent desert lake, lake Turkana. She holds a masters and bachelors degree in fisheries and aquatic science from University of Eldoret, Kenya. Currently she is also a PhD student at the same university in Aquatic Science. Her interests extends from Biogeochemistry of freshwater systems, conservation, and management. She has been involved in various projects such as developing of rapid bioassement tool for monitoring aquatic ecosystems and she is also working as a PhD fellow under the sustainable waterpan project in lower Mara in Kenya and Tanzania. Besides, she also enjoys working with communities in management and conservation of water resources.

*"My current PhD research on the Sustainable Waterpans project in the lower Mara and Serengeti regions requires advanced geospatial analysis, particularly in mapping water pans and assessing their historical changes to increase their sustainability in an ever-changing environment."*

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## Anil Fartiyal

Junior Technical Officer, South Asia- India

Anil is a passionate wildlife enthusiast from India. His academic background in Forestry and Environmental Science, along with over two years of experience in biodiversity conservation, equipped him with the necessary expertise to address biodiversity and environmental challenges. Currently, he is a Junior Technical Officer in the Biodiversity team at Wetlands International South-Asia. Previously, he worked as a Research Biologist for the NRCD (National River Conservation Directorate) project at the Wildlife Institute of India. There, he played a key role in a comprehensive biodiversity assessment of six major Indian rivers. He is a strong proponent of utilizing modern technological advancements to address contemporary wildlife issues. He is constantly seeking innovative solutions to ensure the effective protection of our natural world.

*"This training will allow me to use advanced GIS knowledge that I can integrate with my core specialization."*

## Fidelis Thuo

Student, Cartographay and GIS, Kenya

Fidelis is a student at the Regional Center Training Institute pursuing a diploma in Cartography and GIS. During the course of her program, she has gained skills that she aspires to use to explore environmental issues affecting society today. Fidelis considers herself as a beginner GIS user, but she is highly motivated to advance her skills. She loves exploring new ideas and taking new challenges which has helped her to grow in many ways. She has an experience using GIS software including ArcMap and open source QGIS, and she will continue to seek opportunities to advance her knowledge and grow her career as an environmental GIS specialist.

*"It will help me understand more about GIS and the impact of learning GIS"*

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## Baraka Naftal Mbwambo

Wildlife Research Officer, TAWIRI, Tanzania

Baraka is a wildlife researcher with a background spanning over ten years, driven by a steady passion for assessing wildlife species' habitats to ensure sustainable conservation. Throughout his career, Baraka has groomed his spatial analysis, remote sensing, and data management and analysis skills, enabling him to pinpoint crucial habitats and assess potential threats to wildlife. His dedication lies in harnessing the power of community knowledge and technology to safeguard our planet's biodiversity.

"*This training is an excellent fit for me as it will deepen my expertise in GIS and remote sensing.*"

## Beatrice Chataigner

Wildlife Conservation Consultant, Nairobi, Kenya

Beatrice is an enthusiastic conservationist with 12 years of experience in African protected areas assessment and capacity building. But as to fulfil her deep interest for scientific research, Beatrice recently started a professional reconversion and as a first step, she just completed her MSc in conservation ecology in 2023.

*"This GIS training will be instrumental in helping me processing geospatial data for the manuscript of the scientific paper I am currently developing (building on my MSc thesis) as well as for the PhD project I wish to develop on the influence of climate change on the landscape ecology of the African leopard."*

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## Stella Denis Sanga

Tutorial Assistant, University of Dodoma, Tanzania

Stella graduated from Ardhi University in 2021 with a bachelor's degree in Geoinformatics. She is assisting in teaching remote sensing, introduction to GIS and the geospatial technology applications. Before joining the University, of Dodoma, Stella volunteered at VILAH COMPANY in Mbeya as a Land Surveyor and GIS analyst from November 2021, where she practiced cadastral surveying activities as well as applying Geospatial technology in solving Land related issues.

*"This training will enhance my competence in geospatial technology by deepening my knowledge of geospatial data analysis and applications in analyzing environmental issues through the modules that are expected to be covered. "*

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